

An Automatic Finite-Sample Robustness Metric: Can Dropping a Little Data Make a Big Difference?

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JSM 2026: Data Selection Beyond Estimation Efficiency

¹With coauthors Rafe Meager (UNSW) and Tamara Broderick (MIT)

Dropping data: Mexico Microcredit

Example: Angelucci et al. [2015], a randomized controlled trial study of the efficacy of microcredit in Mexico based on $N = 16,560$ data points. A regression was run to estimate the average effect of microcredit.

Original result: Treatment effect statistically insignificant at 95%.

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Cannot find influential subsets by brute force!

We provide a fast, automatic tool to approximately identify the most influential set of points.

- How do we identify influential sets? When is our method accurate?
- Data dropping as distributional robustness
- Do we identify “the most important datapoints?”

Formalizing the question.

Example: Least squares

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- A particular component: $\hat{\theta}_k$
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Make a qualitative decision using $\phi(\hat{\theta})$ for a smooth, real-valued ϕ .

(WLOG try to increase $\phi(\hat{\theta})$.)

Data dropping as data reweighting.

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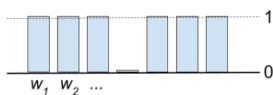
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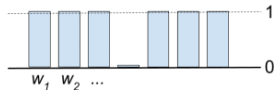
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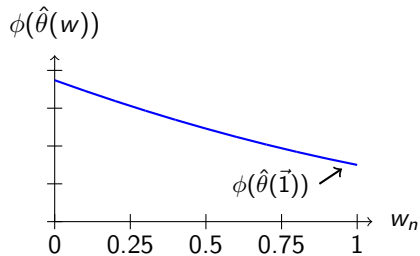
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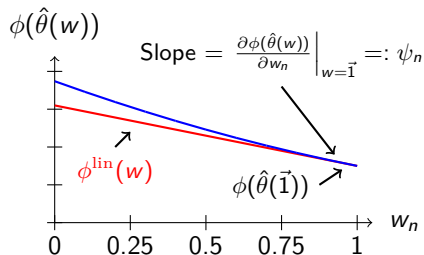


The map $w \mapsto \phi(\hat{\theta}(w))$ is well-defined even for continuous weights.

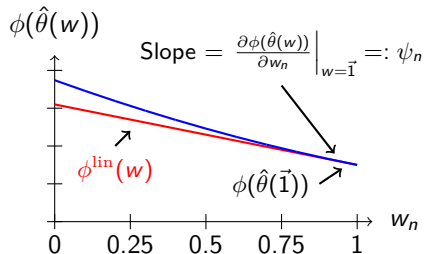
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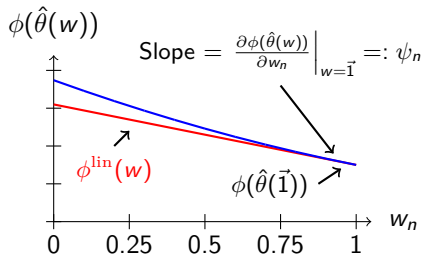


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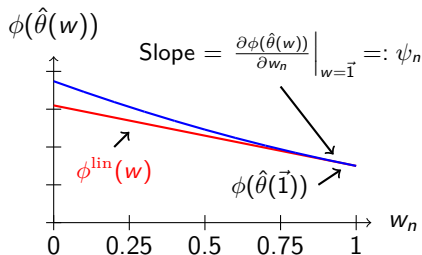


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We can use ψ_n to form a Taylor series approximation:

$$\phi(\hat{\theta}(w)) \approx \phi^{\text{lin}}(w) := \phi(\hat{\theta}(\vec{1})) + \sum_{n=1}^N \psi_n (w_n - 1)$$

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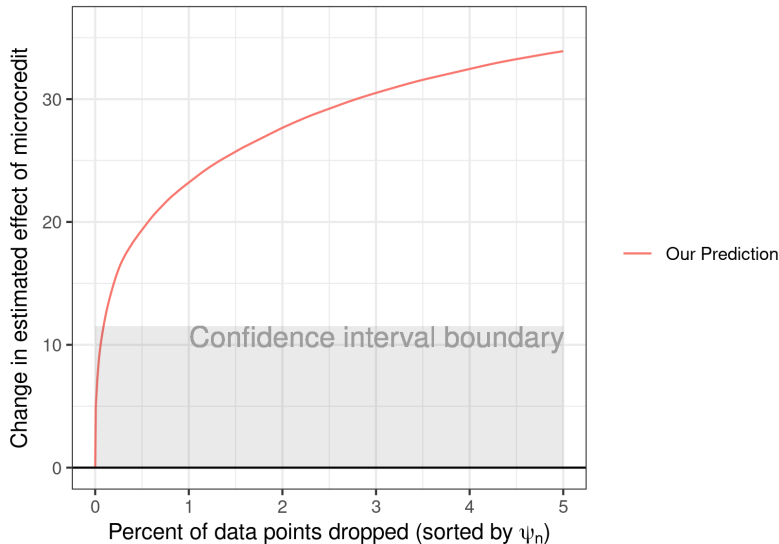
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The most influential points to drop from the linear approximation are easy to identify — they are simply the points with the largest ψ_n !

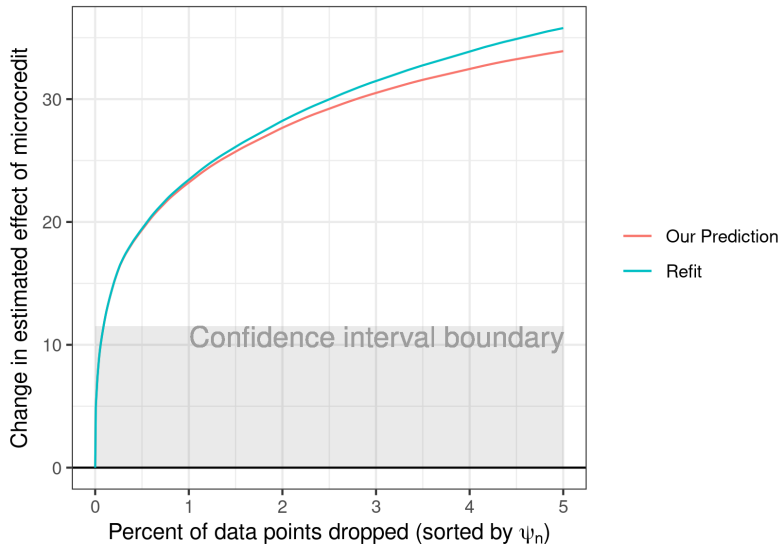
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Conclusion

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- We can quickly and automatically find an approximate influential set which is accurate for small sets.
- Data dropping robustness is principally determined by the signal to noise ratio, and captures sensitivity distinct from sampling and gross error sensitivity.

Links and references

Tamara Broderick, Ryan Giordano, Rachael Meager (alphabetical authors)
“An Automatic Finite-Sample Robustness Metric: Can Dropping a Little Data Change Conclusions?”

<https://arxiv.org/abs/2011.14999>

Select blog posts with more details: <https://rgiordan.github.io>

- Data dropping sensitivity overcomes p-hacking
- Collinearity in OLS after dropping
- Influence functions and sums
- Connections to the bootstrap

Related software on github:

- [rgiordan/zaminfluence](#) (for R)
- [rgiordan/vittles](#) (for Python)

Some of my work on other forms of robustness:

- Prior sensitivity in Bayesian nonparametrics [Giordano et al., 2021]
- Approximate cross-validation (and other reweightings) [Giordano et al., 2019b,a]
- Covariances and prior sensitivity for mean field VB [Giordano et al., 2015, 2018]
- Model sensitivity of MCMC output [Giordano et al., 2018]
- Frequentist variances of MCMC posteriors (in progress)

References

- M. Angelucci, D. Karlan, and J. Zinman. Microcredit impacts: Evidence from a randomized microcredit program placement experiment by Compartamos Banco. *American Economic Journal: Applied Economics*, 7(1):151–82, 2015.
- R. Giordano, T. Broderick, and M. I. Jordan. Linear response methods for accurate covariance estimates from mean field variational Bayes. *Advances in Neural Information Processing Systems*, 28:1441–1449, 2015.
- R. Giordano, T. Broderick, and M. I. Jordan. Covariances, robustness and variational Bayes. *The Journal of Machine Learning Research*, 19(1):1981–2029, 2018.
- R. Giordano, M. I. Jordan, and T. Broderick. A higher-order Swiss army infinitesimal jackknife. *arXiv preprint arXiv:1907.12116*, 2019a.
- R. Giordano, W. Stephenson, R. Liu, M. I. Jordan, and T. Broderick. A Swiss army infinitesimal jackknife. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 1139–1147. PMLR, 2019b.
- R. Giordano, R. Liu, M. I. Jordan, and T. Broderick. Evaluating sensitivity to the stick-breaking prior in Bayesian nonparametrics. 2021.
- F. Hampel. *Robust statistics: The approach based on influence functions*, volume 196. Wiley-Interscience, 1986.